INCH-POUND

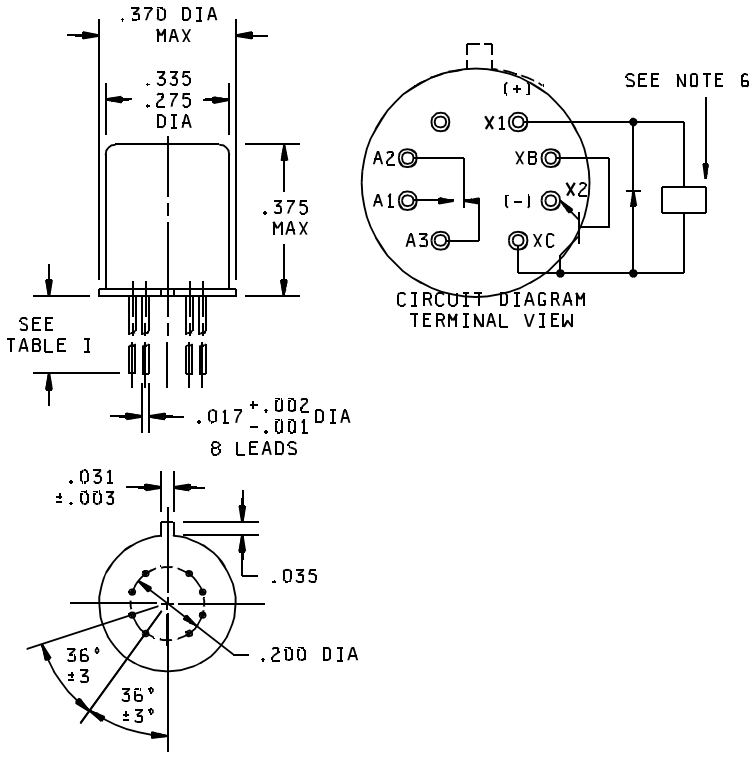
MIL-PRF-28776/4F
8 July 1996
SUPERSEDING
MIL-R-28776/4E
10 February 1984

PERFORMANCE SPECIFICATION SHEET

RELAYS, HYBRID, ESTABLISHED RELIABILITY, SPDT, LOW LEVEL TO 1.0 AMPERE (SENSITIVE COIL OPERATE POWER AT 25°C) (TRANSISTOR DRIVEN) (ELECTROMECHANICAL OUTPUT) WITH DIODE COIL SUPPRESSION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-28776.



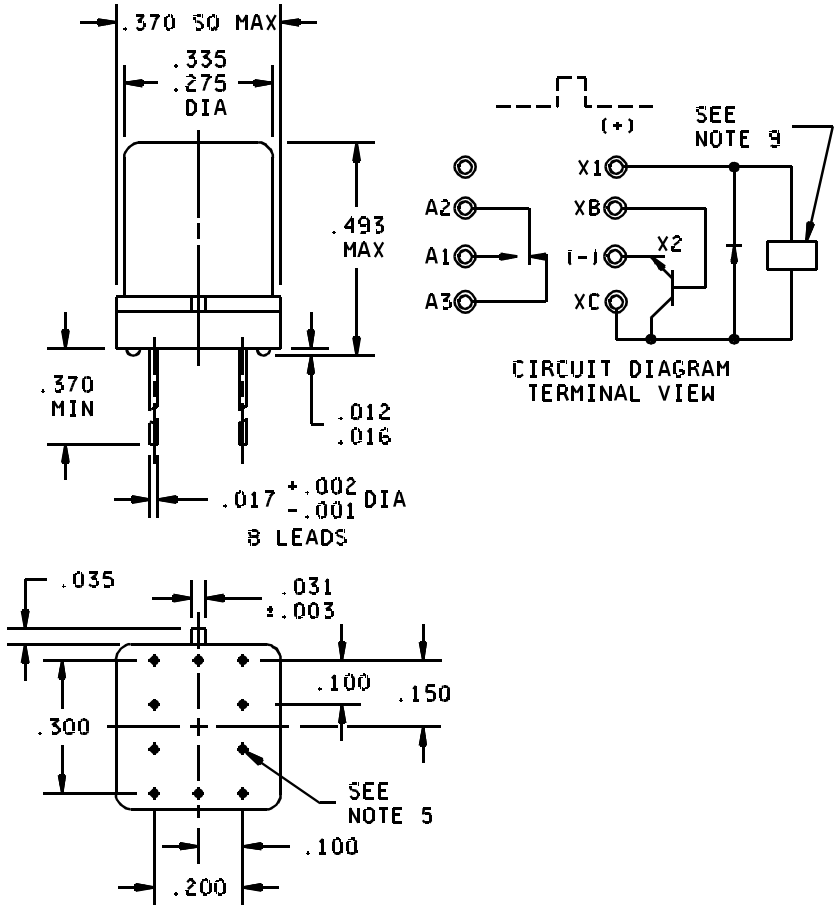
Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.200	5.08
.275	6.98
.335	8.51
.370	9.40
.375	9.52

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.012	0.30
.016	0.41
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.150	3.81
.200	5.08
.275	6.98
.300	7.62
.335	8.51
.370	9.40
.375	9.52
.493	12.52

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to A-A-55485/05-003.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center locations of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Terminal numbers shown above for reference only. Numbers do not appear on relay.
8. Relays shall have a (+) sign placed on circuit diagram as shown.
9. Coil symbol optional in accordance with MIL-STD-1285.
10. Circuit diagram shown on part is the terminal view.

FIGURE 2. Dimensions and configuration relay with spreader pad attached.

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:
1.0 A at 28 V dc.
250 mA at 115 V ac 60 and 400 Hz case not grounded.
100 mA at 115 V ac 60 and 400 Hz case grounded.

Inductive: 0.2 A at 28 V dc with 0.32 H inductance.

Lamp: 0.10 A at 28 V dc.

Low level: 10 µA to 50 µA at 10 mV to 50 mV dc or peak ac maximum.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 100 mΩ maximum, 125 mΩ maximum with spreader pad attached, (tested at 10 mA maximum at 6 V dc maximum or peak ac).

High level:

During life: Not more than 5 percent of open circuit voltage.
After life: 200 mΩ maximum (225 mΩ maximum with spreader pad attached).

Low level:

During life: 33 Ω maximum.
After life: 150 mΩ maximum (175 mΩ maximum with spreader pad attached).

Intermediate current:

During life: 1 Ω maximum.
After life: 200 Ω maximum (225 mΩ maximum with spreader pad attached).

Contact bounce: 1.5 ms maximum.

Contact stabilization time: 2.0 ms maximum.

Overload (high level only): Two times rated current. Not applicable to ac load ratings.

COIL DATA:

Supply voltage: See table I.

Turn-off time: 7.5 ms maximum.

Turn-on time: 3.5 ms maximum.

DIODE DATA:

Coil transient suppression: 1.0 V maximum.

Diode peak inverse voltage: 100 V minimum.

TRANSISTOR DATA:

Specified base turn-off voltage: 0.3 V dc maximum.

Specified base turn-on current: See table I.

Base emitter current: 15 mA maximum.

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ELECTRICAL DATA:

Insulation resistance: 10,000 MΩ minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 MΩ minimum.

Dielectric withstanding voltage:

Sea level:

Between case, frame or enclosure, and all contacts in the energized and deenergized positions: 500 V rms (60 Hz).

Between case, frame or enclosure, and coil: 500 V rms (60 Hz).

Between all contacts and coil: 500 V rms (60 Hz).

Between open contacts in the energized and deenergized positions: 500 V rms (60 Hz).

Altitude:

Between all terminals to case: 125 V rms (60 Hz).

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Shock (specified pulse): Applicable. Test condition B.

Magnetic interference: Applicable.

Salt atmosphere (corrosion): Applicable.

PHYSICAL DATA:

Terminal strength: Applicable.

Pull test: Test condition A, 1 pound pull.
Bend test: Test condition C, 0.5 pound pull.
Twist test: Applicable.

Dimensions and configurations: See figures 1 and 2.

Weight: 3.98 grams (0.14 ounce) maximum, 4.23 grams (0.149 ounce) maximum with spreader pad attached.

Seal: Hermetic.

Minimum marking: As specified in MIL-PRF-28776.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles.
Low level: 100,000 cycles.
Intermediate current: 50,000 cycles.
Mechanical life: 1,000,000 cycles.

Part or Identifying Number (PIN): M28776/4- (dash number from table I and suffix letter designating failure rate level).

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TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/				Supply voltage (V dc) 5/		At +25°C							Over temperature range				
Lead length			Spreader pad (fig 2) 4/			Coil resistance (ref. only) Ohms 6/	Coil circuit current mA 6/ 7/		Spec. Pickup Volt. (V dc) 7/	Spec. Hold-Volt. (V dc) 7/	Spec. Dropout Volt. (V dc) 7/	Spec. base current to assure turn-on (mA)	Spec. Pickup Volt. (V dc) 7/	Spec. Hold Volt. (V dc) 7/	Spec. Drop-out Volt. (V dc) 7/	Spec. base current to assure turn-on (mA)
1.500 min 3/	.187 +.050, -.010	.500 min		Rated	Max		Max	Min								
017	025	033	041	5.0	8.0	125	47.8	34.7	2.6	1.4	0.22	.28	3.6	2.0	0.15	1.20
018	026	034	042	6.0	11.0	255	27.7	21.2	3.5	2.0	0.28	.20	4.8	2.8	0.18	0.78
019	027	035	043	9.0	16.0	630	16.8	11.8	5.4	3.0	0.54	.13	7.8	4.2	0.35	0.48
020	028	036	044	12.0	22.0	1,025	13.6	10.1	6.6	4.0	0.63	.10	10.0	5.6	0.41	0.39
021	029	037	045	18.0	33.0	2,300	9.1	6.7	9.8	6.0	0.91	.07	14.5	8.4	0.58	0.26
022	030	038	046	26.5	45.0	4,000	7.7	5.7	12.8	8.0	1.37	.05	19.0	10.4	0.89	0.20
023	031	039	047	32.0	57.0	6,500	5.8	4.2	18.7	10.0	1.60	.04	24.0	15.0	1.00	0.16
024	032	040	048	40.0	75.0	11,000	4.3	3.1	23.3	13.0	2.10	.03	30.0	18.7 R	1.30 R	0.13

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example 017L - - - - - 032R.

3/ 1.500 leads are inactive for new design.

4/ Relays supplied with spreader pads (-041 through -048) shall have the pad rigidly attached.

5/ CAUTION: The use of any supply voltage less than the rated supply voltage will compromise the operation of the relay.

6/ When rated supply voltage is applied to the supply terminals, the coil circuit current must be within the limits shown. Measure at 25 at rated supply voltage for 5 seconds, maximum.

7/ Set base current from 3 mA to 15 mA during electrical measurements.

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QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
37 units plus 1 unsealed unit for level L at C = 1 2/ 63 units plus 1 unsealed unit for level M at C = 1 2/ Qualification inspection as applicable.	M28776/4-038	37 units plus 1 unsealed unit for level L at C = 1 2/ 63 units plus 1 unsealed unit for level M at C = 1 2/ Qualification inspection as applicable.
	M28776/4-033 M28776/4-034 M28776/4-035 M28776/4-036 M28776/4-037 M28776/4-039 M28776/4-040	2 units each PIN. Qualification inspection, group I

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-PRF-28776/1, MIL-PRF-28776/3, and MIL-PRF-28776/5 may be used in addition to MIL-PRF-28776/4 data.

2/ The number of units required for qualification testing will be increased as required in group Q5 of table II, MIL-PRF-28776, if the relay manufacturer elects to test the number of units permitting two or more failures. Prior to performance of qualification inspection testing, the relay manufacturer shall preselect the sampling plan.

Qualification inspection testing for relays supplied with spreader pads (-041 through -048), two units of the 26.5 volt rated coil voltage (-046) shall be tested as follows:

Perform A1 tests of group A inspection. Relay leads shall be formed and the mounting pad removed. Perform seal test. Rigidly attach mounting pad to relay. Perform A2 and A3 tests of group A inspection. Perform visual and mechanical inspection (external).

Group A inspection testing for relays supplied with spreader pads (-041 through -048) shall be tested as follows:

Perform A1 tests of group A inspection. Relay leads shall be formed and the mounting pad removed. Perform seal test. Rigidly attach mounting pad to relay. Perform A2 and A3 tests of group A inspection. Perform visual and mechanical inspection (external).

Qualification inspection (reduced testing) and sample size: See table III. If the relays produced for MIL-PRF-28776/4 are similar in construction and design except for the coils to the relays produced for MIL-PRF-28776/3, then reduced testing for qualification of MIL-PRF-28776/4 relays may be performed concurrent with or subsequent to successful qualification of MIL-PRF-28776/3 relays.

TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each supply voltage: Group Q1 of qualification inspection table. 1 unsealed sample unit: Internal examination

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SUPERSESSION DATA:

Supersession data: See table IV.

TABLE IV. Supersession data.

Superseded PIN M28776/4-	New PIN M28776/4-
001	017
002	018
003	019
004	020
005	021
006	022
007	023
008	024
009	025
010	026
011	027
012	028
013	029
014	030
015	031
016	032

1/ Dash numbers -017 through -024 are inactive for new design and are for support of existing equipment designs only.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, ME, MI
Air Force - 17, 19, 99
Navy - AS, OS, SH
NSA - NS

Preparing activity:
DLA - CC

(Project 5945-0967-03)

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